Caption Binding Ambiguity in Academic Document Parsing

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Abstract

This fixture tests whether parsers correctly bind captions to their parent elements when a table and a figure appear in close proximity with similar-looking caption text. The captions below intentionally use parallel structure: both start with 'Overview of' followed by a description of their respective content.

1. Introduction

Academic documents often place tables and figures in close proximity, especially in results sections. This creates ambiguity for automated parsers that must decide which caption belongs to which element. We present a controlled fixture that exercises this ambiguity.

The following table and figure present our experimental setup and a visualization of the data distribution. Both elements carry captions with parallel structure to maximize binding ambiguity.

Dataset	Samples	Classes
MNIST	70000	10
CIFAR-10	60000	10
ImageNet	1.2M	1000

Overview of the dataset statistics used in our evaluation.

Overview of the data distribution across training and test splits.

2. Analysis

The key challenge in caption binding is determining whether 'Overview of the dataset statistics' belongs to the table and 'Overview of the data distribution' belongs to the figure, or vice versa. A robust parser should use element type cues (table vs. figure) and proximity to make the correct assignment.

In this fixture, the table appears first with its caption immediately below, followed by the figure with its caption. The parallel caption structure is the intended ambiguity signal.

3. Additional Ambiguity Case

We include a second case with reversed ordering: figure first, then table, again with parallel captions.

Summary of the model architecture components.

Component	Parameters	Layer Type
Encoder	85M	Transformer
Decoder	92M	Transformer
Classifier	1.2M	Linear

Summary of the training hyperparameters and optimization settings.

4. Conclusion

This fixture provides a controlled test for caption binding ambiguity resolution. Parsers should be able to distinguish between table and figure captions even when they share similar sentence structure.